Linux History and Operation

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1. The Evolution of Linux

What is Linux?

- Linux is a **free and open-source operating system**
- It works like Unix but it is **completely free**
- Anyone can use, modify, and share it

How did it start?

```
UNIX (1969)

MINIX (1987) - educational OS

Linux (1991) - by Linus Torvalds
```

The Story

- **1969** Unix was created at AT&T Bell Labs
  - It was powerful but **not free**
- **1987** Professor Tanenbaum made **MINIX**
  - It was for students to learn OS concepts
  - But you could **not modify** it freely
- **1991** **Linus Torvalds** (21 year old student from Finland)
  - Was not happy with MINIX limitations
  - Started writing his **own kernel**
  - Posted online: *"I am making a free OS, just a hobby..."*
  - People loved it and started helping him
  - This became **Linux**

How Linux Grew Over Time

Year	What Happened
1991	Linux kernel version 0.01 released
1992	Made open source under GPL license
1994	Linux version 1.0 released
1998	Big companies like IBM started using Linux
2003	Linux 2.6 - became very powerful
2008	Android (based on Linux) launched
Today	Powers 90%+ of internet servers

Where is Linux used today?

- **Web Servers** - Google, Facebook, Amazon all run on Linux
- **Android Phones** - Android is built on Linux kernel
- **Supercomputers** - Top 500 supercomputers use Linux
- **Cloud Computing** - AWS, Azure use Linux
- **Desktops** - Ubuntu, Fedora, Mint

Simple Summary

> *Linux started as a student hobby project in 1991. Over time, millions of developers contributed to it. Today it is the most used operating system in the world for servers and technology.*

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2. The GNU Movement and the GPL

Background - Why was GNU needed?

- In the 1980s, software was **expensive and closed**
- You could not see how it worked inside
- You could not modify or share it
- **Richard Stallman** was angry about this

What is GNU?

```
GNU = "GNU's Not Unix"
(It is a recursive acronym - funny name)
```

- Started in **1983** by **Richard Stallman** at MIT
- Goal was to create a **completely free operating system**
- GNU provided all the important tools:

GNU Tool	What it does
`gcc`	Compiler - converts code to program
`bash`	Shell - command line interface
`glibc`	C Library - basic functions
`make`	Builds software automatically
`tar, grep`	File and text tools

The Free Software Foundation (FSF)

- Stallman created **FSF in 1985**
- They defined what “free software” means
- Free does NOT mean free of cost
- Free means **freedom to use, study, modify, share**

4 Freedoms of Free Software

Freedom	Meaning
Freedom 0	Run the program for any purpose
Freedom 1	Study how it works (see source code)
Freedom 2	Share copies with others
Freedom 3	Modify and share your improved version

The Problem GNU Had

- GNU had all the tools
- GNU did NOT have a working kernel

- Linux had a working kernel
- **GNU + Linux = Complete Operating System**

```
GNU Tools + Linux Kernel = GNU/Linux (Complete OS)
```

What is GPL?

GPL = General Public License

- A legal license created by Stallman in **1989**
- Protects software freedom using law
- Based on concept called **Copyleft**

Copyright vs Copyleft

```
Normal Copyright:
"You cannot copy, modify, or share this"

Copyleft (GPL):
"You CAN copy, modify, and share this
BUT you must keep it free for others too"
```

GPL Rules - Simple Version

- You can use the software freely
- You can see the source code
- You can modify it
- You can share it
- If you share it, you MUST share the source code
- You MUST keep the same GPL license

GPL Versions

Version	Year	Key Addition
GPLv1	1989	Basic copyleft rules
GPLv2	1991	Cannot distribute if you remove freedoms
GPLv3	2007	Added protection against patents

> *Linux Kernel uses **GPLv2** license*

Simple Summary

> *GNU movement gave us free tools and the GPL license. The GPL makes sure that free software stays free forever. When Linux kernel combined with GNU tools, we got a complete free operating system that anyone can use.*

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3. Linux Operations as a Server

Why Do Companies Choose Linux for Servers?

Reason	Explanation
Free	No license cost, saves millions of dollars
Stable	Runs for years without crashing or restarting
Secure	Strong security model, fewer viruses
Fast	Uses resources efficiently
Flexible	Can be customized for any need
Support	Huge community + paid enterprise support available

Popular Linux Server Distributions

Distribution	Best For
Ubuntu Server	General use, cloud, beginners
Red Hat Enterprise Linux	Large enterprises, paid support
CentOS / Rocky Linux	Free alternative to Red Hat
Debian	Very stable, long support period
SUSE Linux Enterprise	European enterprises

Types of Servers Linux Can Run

Web Server

- Hosts websites on the internet
- Software used: **Apache, Nginx**
- Example: When you visit a website, Linux server sends you that page

File Server

- Stores and shares files across a network
- Software used: **Samba** (for Windows), **NFS** (for Linux)
- Example: Shared drives in an office

Mail Server

- Sends and receives emails
- Software used: **Postfix, Sendmail, Dovecot**
- Example: Company email system

Database Server

- Stores and manages data
- Software used: **MySQL, PostgreSQL, MariaDB**
- Example: Stores user accounts, product data

DNS Server

- Converts website names to IP addresses
- Software used: **BIND, dnsmasq**
- Example: google.com 142.250.80.46

Firewall Server

- Protects the network from attacks
- Software used: **iptables, firewalld**
- Example: Blocks unauthorized access

Cloud / Virtual Server

- Runs multiple virtual machines
- Software used: **KVM, Docker, Kubernetes**
- Example: AWS EC2 instances

Important Linux Server Commands

```
# Check system info
uname -a            # Shows kernel version
uptime              # How long server has been running
hostname            # Server name

# Manage services
systemctl start nginx      # Start a service
systemctl stop nginx       # Stop a service
systemctl restart nginx    # Restart a service
systemctl status nginx     # Check if running

# Check resources
top                 # See CPU and memory usage
df -h               # Check disk space
free -h             # Check memory

# Remote access
ssh user@serverIP     # Connect to server remotely

# Check network
ip addr             # Show IP addresses
netstat -tulnp      # Show open ports
```

How Linux Server is Managed?

```
Admin's Computer

   SSH (Secure Connection over Internet)

Linux Server (anywhere in the world)

Run commands to manage everything
```

Linux Server Security Tips

- Use **SSH keys** instead of passwords
- Set up **firewall rules**
- Keep system **updated regularly**

- **Monitor logs** for suspicious activity
- Use **fail2ban** to block hackers

Simple Summary

> *Linux is the number one choice for servers because it is free, stable, and secure. It can act as a web server, database server, file server, and much more. System administrators manage Linux servers remotely using SSH and command line tools.*

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4. Architecture and Structure of Linux

Basic Architecture - Big Picture

```
      USER
(You - using applications)

    APPLICATIONS
(Browser, editor, games)

      SHELL
(Takes your commands)

      KERNEL
(Brain of the OS)

     HARDWARE
(CPU, RAM, Disk, Network)
```

Layer by Layer Explanation

Layer 1 - Hardware (Bottom)

- Physical parts of the computer
- CPU, RAM, Hard Disk, Network Card, Monitor
- Linux kernel directly controls these

Layer 2 - Kernel (Most Important)

Kernel = The Brain of Linux

- It is the core part of the OS
- Manages everything between software and hardware
- Users cannot directly touch kernel - it is protected

What does Kernel do?

Job	Explanation
Process Management	Decides which program runs and when
Memory Management	Gives RAM to programs, takes it back when done
File System	Manages how files are stored and read
Device Drivers	Communicates with hardware devices
Network Management	Handles internet and network connections
Security	Controls who can access what

Kernel Space vs User Space

```
USER SPACE
(Your programs run here)
(Limited access - cannot touch hardware directly)
        System Calls (requests to kernel)
KERNEL SPACE
(Kernel runs here)
(Full access to everything)
(Controls all hardware)
```

Linux Kernel Type

- Linux uses **Monolithic Kernel**
- Means most functions are inside one big kernel
- But you can add/remove **modules** (extra features) as needed

```
lsmod         # See loaded modules
modprobe wifi  # Load wifi module
rmmod wifi     # Remove wifi module
```

Layer 3 - Shell (Interface)

Shell = The Translator between You and Kernel

- You type a command
- Shell understands it
- Shell tells the kernel what to do
- Kernel does the work
- Shell shows you the result

Types of Shells

Shell	Description
bash	Most common, default in most Linux
sh	Original Unix shell
zsh	Advanced shell with extra features
fish	User friendly, colorful shell

Example of How Shell Works

```
You type:   ls -l

Shell understands: "List files with details"

Shell asks Kernel: "Please read directory"

Kernel reads files from disk

Shell shows you the result on screen
```

Layer 4 - File System Structure

In Linux - Everything is a File

- Programs are files
- Devices are files
- Settings are files

Linux Folder Structure (FHS - Filesystem Hierarchy Standard)

```
/     Root (top level, like C:\ in Windows)

      /home      User personal files (like My Documents)
      /etc       All configuration files (settings)
      /bin       Essential commands (ls, cp, mv)
      /sbin      Admin commands (for root user)
      /var       Log files, database files
      /tmp       Temporary files
      /dev       Device files (hard disk, USB)
      /proc      Running process information
      /boot      Boot files and kernel
      /usr       Programs and software
      /lib       System libraries
      /root      Home folder for admin (root user)
```

Important Folders Explained Simply

Folder	What is stored there
`/home/username`	Your personal files and settings
`/etc`	Config files - like settings files of all programs
`/var/log`	All log files - helps find errors
`/dev/sda`	Your hard disk is stored as a file here
`/proc/cpuinfo`	Read this file to see CPU details
`/tmp`	Temporary files - deleted on reboot

Layer 5 - Applications (Top)

- Programs that users run

- Examples: Firefox, Apache web server, MySQL
- These use Shell and System Calls to talk to Kernel

The Boot Process - How Linux Starts

```
Step 1: BIOS/UEFI
        (checks hardware, finds boot disk)

Step 2: Bootloader (GRUB2)
        (loads the Linux kernel from /boot)

Step 3: Kernel starts
        (initializes hardware, mounts root filesystem)

Step 4: systemd starts (PID 1)
        (first process, starts all other services)

Step 5: Services start
        (web server, database, network, etc.)

Step 6: Login screen appears
        (ready for user)
```

Process Management

- Every running program is called a **Process**
- Each process gets a unique **PID (Process ID)**
- **systemd** is always PID 1 (first process)

```
ps aux       # Show all running processes
top          # Live view of processes
kill 1234    # Stop process with PID 1234
```

Process States

```
New    Running    Waiting    Sleeping    Terminated
```

Device Files in Linux

- Linux shows hardware as files in `/dev/` folder

Device File	What it represents
`/dev/sda`	First hard disk
`/dev/sdb`	Second hard disk
`/dev/tty`	Terminal (keyboard/screen)
`/dev/null`	Trash bin (data sent here is deleted)
`/dev/random`	Generates random numbers

Quick Revision - All Topics

Topic	Key Points to Remember
Evolution	1991, Linus Torvalds, Finland, started as hobby, now global
GNU	1983, Richard Stallman, free tools for OS
GPL License	Free to use/modify/share BUT keep it open source
Linux as Server	Free, stable, secure, runs web/DB/file/DNS servers
Kernel	Brain of OS, manages CPU/RAM/devices
Shell	Interface between user and kernel
File System	Everything is a file, starts from / (root)
Boot Process	BIOS GRUB Kernel systemd Login

Important Questions

Q1. Who created Linux and when?

> *Linus Torvalds created Linux in 1991 when he was a student in Finland.*

Q2. What is the difference between Linux and Unix?

> *Unix is old, expensive, and proprietary. Linux is free, open-source, and based on Unix ideas but written from scratch.*

Q3. What is GNU and why is it important?

> *GNU is a project started by Richard Stallman in 1983 to create free software tools. The Linux kernel combined with GNU tools gives us a complete free operating system.*

Q4. What is GPL license?

> *GPL is a free software license that allows anyone to use, modify, and share software. But if you share it, you must also share the source code and keep the same license.*

Q5. Why do companies use Linux for servers?

> *Because Linux is free, stable, secure, and performs well. It can run for years without needing a restart.*

Q6. What is the Linux kernel?

> *The kernel is the core of the operating system. It manages CPU, memory, devices, and communication between hardware and software.*

Q7. What is a shell in Linux?

> *Shell is an interface that takes commands from the user, passes them to the kernel, and shows the results. Bash is the most common shell.*

Q8. Explain Linux file system structure.

> *Linux uses a tree structure starting from root (/). Important folders are /home for users, /etc for settings, /var for logs, and /boot for boot files.*

Q9. What is systemd?

> *systemd is the first process that starts in Linux (PID 1). It starts all other services and manages the system.*

Q10. What do you mean by everything is a file in Linux?

> *In Linux, not just documents but also hardware devices, processes, and network connections are represented as files. This makes it easy to manage everything using same file commands.*